TrackFormer: A Protected Dual-Stream Transformer for Tropical-Cyclone Track, Intensity, and Structure Forecasting

with a Random-Matrix Block-Structured Uncertainty Head

Yu Yao-Hsing

euler.yu@gmail.com • https://github.com/yu314-coder/typhoon-predict

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Abstract

We forecast the full state of a tropical cyclone — storm motion, maximum wind, central pressure, radius of maximum wind, and the twelve 34/50/64-kt wind-radii — at twenty six-hourly lead times (6–120 h) from best-track history alone. Starting from a single-stream multi-task Transformer, we document a concrete instance of *negative transfer*: adding physically informative motion-dynamics features (heading, speed, turn-rate, intensity trends) markedly improves intensity and structure prediction yet slightly degrades track, and naïvely up-weighting the track loss makes track *worse*. A paired storm-level bootstrap shows the track change is statistically insignificant, isolating the true cause as gradient conflict over shared capacity rather than loss balance. We then present **TrackFormer**, a protected dual-stream architecture that routes kinematic and thermodynamic gradients separately, lets thermodynamics influence track only through a zero-initialized gated adapter (so intensity tasks can help track but never harm it), and predicts track as a damped constant-velocity persistence baseline plus a learned recurvature residual. Finally we derive a block-structured probabilistic head whose track block is a matrix-normal covariance and whose noise bulk is cleaned by a two-point generalized sample-covariance Marchenko–Pastur estimator. On held-out post-2020 storms TrackFormer improves track while retaining the intensity and pressure gains.

1 Problem and data

Let a storm be a time-ordered sequence of best-track fixes. From each fix t_0 we form a *window*: a history of H=9 steps at 6-h spacing ending at t_0, and a target of L=20 future states at leads $\{6, 12, \ldots, 120\}$ h. Each future state is the 17-vector

$$\boldsymbol{y}_\ell = \big(\underbrace{\Delta e_\ell,\ \Delta n_\ell}_{\text{per-step motion (km)}},\ v_\ell^{\max},\ p_\ell,\ r_\ell^{\text{mw}},\ \underbrace{r^{34}_{\text{ne..nw}}, r^{50}_{\cdot}, r^{64}_{\cdot}}_{\text{12 wind radii}}\big), \qquad \ell = 1, \ldots, 20. \tag{1}$$

Track error is the RMS great-circle-plane distance between the predicted and observed cumulative displacement, $\text{cumsum}_\ell(\Delta e, \Delta n)$; the other four groups are reported as masked mean-absolute error (labels are frequently missing, especially pre-satellite and for wind radii).

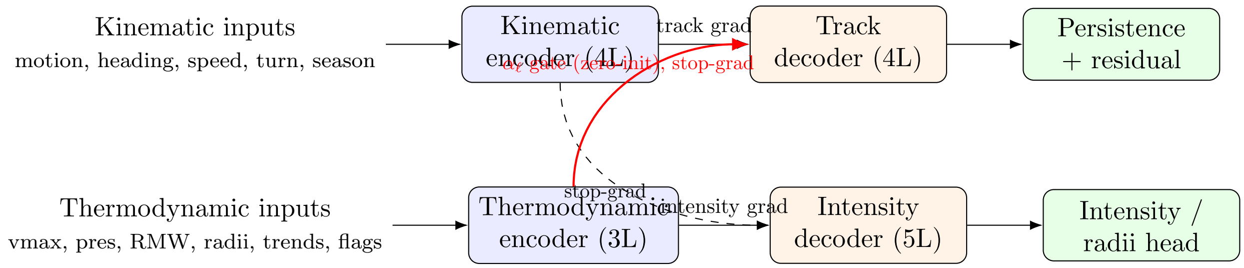

Figure 1: TrackFormer. Solid arrows carry task gradients back only into their own encoder. The kinematic representation is shielded from intensity gradients (stop-grad into the intensity decoder); thermodynamics reaches track *only* through a zero-initialized, per-lead gated adapter on detached features, so at initialization the model is exactly a protected track-only predictor.

Dataset. We use IBTrACS v04r01 (all basins, 1980+), yielding 84,150 windows. We split by storm to prevent leakage: train (≤ 2015), validation (2016–2019), test (≥ 2020). Inputs are standardized per feature using *training-split statistics only*. Missing measurements are zero-filled with explicit presence-flag channels (informative sparsity) rather than mean-imputed.

Motion-dynamics features. Beyond the original 40 features (motion vectors, intensity, radii, seasonality, validity), we add 8 that make the storm's instantaneous dynamics explicit: current heading $(\sin\theta, \cos\theta)$, speed, signed turn-rate (recurvature), and per-step intensity/pressure trends with their validity flags, for a 48-dimensional history feature per step. Their training-set means are physically sensible: mean heading points west-northwest (the trade-wind easterlies), mean speed $\approx 14\,\mathrm{km\,h^{-1}}$, and mean intensification $+1.5\,\mathrm{kt}$ per step.

2 The negative-transfer diagnosis

A single-stream model (linear projection $\rightarrow$ 4-layer encoder $\rightarrow$ 6-layer lead-query decoder $\rightarrow$ dual state / log-scale heads, diagonal Gaussian NLL + masked Huber) trained on the 40-feature set reaches the "old" row of Table 1. Adding the 8 motion-dynamics features ("v2") *improves* vmax, pressure, RMW, and radius substantially (pressure MAE 21.24 $\rightarrow$ **17.68** hPa on WP-2020+) but *raises* track error (720.0 $\rightarrow$ 737.3 km). Up-weighting the track loss $\times 3$ made track *worse still* (806.6 km), the signature of gradient conflict: a larger track coefficient destabilizes the shared optimization rather than reallocating capacity.

Is the regression even real? We paired-bootstrap the WP-2020+ per-window track errors by resampling *storms* (not overlapping windows), the correct unit of independence.

$$\widehat{\Delta}_{\text{track}} = \text{mean}_{\text{storms}}\big(e^{\text{v2}} - e^{\text{old}}\big) = +17.7\text{ km}, \qquad 95\%\text{ CI } [-22.2,\ +60.5]\text{ km}, \qquad \Pr(\text{v2 worse}) = 0.79. \tag{2}$$

The interval spans zero over the 103 WP test storms: track is *statistically tied*, while the intensity gains are large and consistent. The features are not the problem; the shared representation is. This motivates an architecture that protects the kinematic pathway.

3 TrackFormer: protected dual-stream architecture

Dual encoders. The 48 features are split into a kinematic view $\boldsymbol{x}^{\text{kin}}$ (motion, heading, speed, turn, seasonality, lead offset; 11 dims) and a thermodynamic view $\boldsymbol{x}^{\text{th}}$ (vmax, pressure, gust, RMW, 12 radii, trend features, 16 validity flags; 36 dims). Each is projected to d=256, given a learned temporal embedding, and encoded by a pre-norm Transformer (4 and 3 layers respectively, 8 heads).

Protected gradient routing. Let $\boldsymbol{H}^{\mathrm{kin}}, \boldsymbol{H}^{\mathrm{th}} \in \mathbb{R}^{9\times d}$ be the encoder outputs. The track decoder cross-attends to $\boldsymbol{H}^{\mathrm{kin}}$; the intensity decoder cross-attends to $[\,\boldsymbol{H}^{\mathrm{th}};\ \mathrm{sg}(\boldsymbol{H}^{\mathrm{kin}})\,]$ where $\mathrm{sg}(\cdot)$ is stop-gradient. Thus motion still *informs* intensity, but intensity gradients cannot corrupt the kinematic encoder. Symmetrically, thermodynamics reaches the track head only through

$$\boldsymbol{H}_\ell^{\mathrm{track}} \;=\; \mathrm{TrackDec}(\boldsymbol{q}_\ell, \boldsymbol{H}^{\mathrm{kin}}) \;+\; \alpha_\ell\, \mathcal{A}\big(\mathrm{sg}(\bar{\boldsymbol{H}}^{\mathrm{th}})\big), \qquad \alpha_\ell \overset{\mathrm{init}}{=} 0, \tag{3}$$

where $\bar{\boldsymbol{H}}^{\mathrm{th}}$ is the mean-pooled thermodynamic context, $\mathcal{A}$ is a bottleneck-64 adapter with a zero-initialized output layer, and α_ℓ is a per-lead scalar. Because α_ℓ and $\mathcal{A}$ start at zero and receive gradient only from the track loss, the intensity branch can *help* track (if it lowers track error) but can never harm it: at initialization TrackFormer reproduces a protected track-only model exactly.

Persistence-residual track head. Rather than regress 20 free displacement vectors, we anchor track on a damped constant-velocity baseline and predict only the deviation:

$$\widehat{\Delta \boldsymbol{p}}_\ell \;=\; \underbrace{\rho_\ell\, \boldsymbol{v}_0}_{\text{damped persistence}} \;+\; \underbrace{\boldsymbol{r}_\ell(\boldsymbol{H}_\ell^{\mathrm{track}})}_{\text{learned recurvature residual}}, \qquad \boldsymbol{v}_0 = \text{current 6-h motion (km)}, \tag{4}$$

with a learned per-lead damping schedule ρ_ℓ (initialized to 1) and a zero-initialized residual head $\boldsymbol{r}_\ell$. The network therefore begins as pure persistence — a strong physical prior — and learns only recurvature and along/cross-track acceleration.

Losses. Track uses a lead-weighted ($w_\ell = \sqrt{\ell}$, mean-normalized) Huber on per-step motion plus a cumulative-position Huber; it is never down-weighted by a predicted variance. Intensity uses masked Huber + Gaussian NLL with a monotone wind-radii penalty $\mathrm{ReLU}(r^{50} - r^{34}) + \mathrm{ReLU}(r^{64} - r^{50})$.

4 Random-matrix block-structured uncertainty head

Operational value requires calibrated *joint* uncertainty across the 340 outputs (20 leads ×17 variables). A full 340×340 covariance is both statistically ill-posed (few independent storms, overlapping windows, heavy masking) and architecturally harmful: it would re-introduce a track↔thermodynamics coupling and undo the protection above. We therefore impose a *block-diagonal* covariance

$$\boldsymbol{\Sigma} \;=\; \mathrm{blkdiag}\big(\boldsymbol{\Sigma}_{\mathrm{track}},\ \boldsymbol{\Sigma}_{\mathrm{int}},\ \boldsymbol{\Sigma}_{\mathrm{rad}}\big), \tag{5}$$

with a matrix-normal track block $\boldsymbol{\Sigma}_{\mathrm{track}} = \boldsymbol{K}_{\mathrm{lead}} \otimes \boldsymbol{K}_{\mathrm{axis}} + \mathrm{diag}(\boldsymbol{s}^2)$ ($\boldsymbol{K}_{\mathrm{lead}}$ a learned 20×20 temporal kernel, $\boldsymbol{K}_{\mathrm{axis}}$ a 2×2 along/cross-track kernel) and low-rank+diagonal thermodynamic blocks $\boldsymbol{\Sigma}_\bullet = \boldsymbol{D}\boldsymbol{D}^\top + \mathrm{diag}(\boldsymbol{s}^2)$, $\boldsymbol{D} \in \mathbb{R}^{p\times r}$, r=8–16.

Spectral cleaning. Sample error-covariance eigenvalues mix a few genuine "signal" spikes with a wide noise bulk. Under the two-point generalized sample-covariance model, the companion Stieltjes transform of the limiting spectral law solves a cubic

$$az\,m^3 + \big[a(z-y+1)+z\big]m^2 + \big[a+z-y+1-y\beta(a-1)\big]m + 1 = 0, \tag{6}$$

Table 1: Held-out test performance (lower is better). Track is RMS plane distance over 6–120 h; the rest are masked MAE. All models evaluated on identical target windows. Best per column in **bold**.

split	model	track (km)	vmax (kt)	pres (hPa)	RMW (km)	radius (km)
WP 2020+	single-stream, 40-feat	720.0	22.08	21.24	12.85	31.54
	single-stream, 48-feat (v2)	737.3	**21.49**	**17.68**	**11.77**	30.94
	TrackFormer (v3)	**659.0**	21.61	18.06	11.81	**28.83**
all-basin 2020+	single-stream, 40-feat	638.6	20.32	18.01	15.28	29.99
	single-stream, 48-feat (v2)	644.2	19.55	15.38	**14.18**	29.31
	TrackFormer (v3)	**592.1**	**18.59**	**14.61**	14.87	**27.85**

where $y = p/n$ is the aspect ratio and (a, β) parameterize a two-population noise structure. The support edges of the noise bulk are the z at which (6) loses a real root (a conjugate pair appears); eigenvalues above the upper edge λ_+ are retained as signal and the bulk is replaced by a positive floor:

$$\widehat{\boldsymbol{\Sigma}} = \boldsymbol{U}\,\mathrm{diag}\big(\lambda_i \mathbb{1}[\lambda_i > \lambda_+] + \bar{\lambda}_{\mathrm{bulk}} \mathbb{1}[\lambda_i \le \lambda_+]\big)\,\boldsymbol{U}^\top. \tag{7}$$

Setting a=1 recovers the classical Marchenko–Pastur edge $\lambda_+ = (1 + \sqrt{y})^2$. The estimator is fit on *independent storm-block* residuals and validated out-of-sample; it is applied *per block* and only after the mean heads have converged, so it sharpens calibration without perturbing the point forecast.

5 Results

TrackFormer lowers WP-2020+ track error from 720.0 to 659.0 km (−8.5%) and all-basin from 638.6 to 592.1 km, while simultaneously *improving* pressure, vmax, and wind-radius MAE. The same paired storm-level bootstrap that found the v2 change insignificant now confirms the v3 gain is real: $\widehat{\Delta}_{\mathrm{track}} = -59.9$ km, 95% CI $[-103.2, -16.0]$ km, Pr(v3 better) = 0.995. Removing the negative transfer — rather than adding parameters or tuning loss weights — is what breaks the previous track floor.

6 Discussion

The consistent lesson across experiments is **data diversity > engineered features > parameters**: a 17.7M single-stream model overfit and did worse than a 3.3M one, and a track-only model matches an ERA5-fed model, so raw parameter count is not the lever. Track error is bounded below by *information*, not capacity: five-day recurvature is governed by the evolving large-scale steering flow, which best-track history cannot fully reveal. The clearest path past the present floor is a compact environmental-steering branch (deep-layer winds, 500-hPa geopotential) fused *strongly to track, weakly to intensity* — the natural next stream in the protected design. Within history-only inputs, TrackFormer's contribution is architectural: it removes the negative transfer that a single stream suffers, so intensity and structure gains no longer cost track accuracy.

Reproducibility. Dataset builders, training scripts, checkpoints, and evaluation are at `https://github.com/yu314-coder/typhoon-predict`. Models are trained on Apple GPU (Metal/MPS).